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Discover Mechanical Engineering

REVIEW Open Access

Advances in thermoplastic composites focusing on processing, recyclability, and performance

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Abstract

Thermoplastic composites have emerged as advanced engineering materials due to their superior recyclability, damage tolerance, and processing versatility compared with conventional thermoset systems. This review provides a comprehensive assessment of recent advances in thermoplastic composite materials, with particular emphasis on matrix and reinforcement selection, advanced processing technologies, recyclability, and performance-related challenges. Key manufacturing approaches such as compression molding, thermoforming, automated tape placement, and in-situ consolidation are critically discussed, highlighting their advantages, limitations, and influence on microstructural integrity and mechanical performance. The review further examines sustainability aspects through life cycle assessment, renewable feedstocks, and end-of-life management strategies, positioning thermoplastic composites as promising candidates for circular economy-driven applications. Despite these advantages, long-term durability issues remain significant, particularly under dynamic loading, environmental exposure, and hygrothermal conditions. Challenges related to viscoelastic damage evolution, interfacial bonding, moisture sensitivity, and residual stress development are also discussed. Finally, the review identifies existing research gaps and outlines future directions focused on interfacial engineering, advanced process control, data-driven modeling, and sustainable material design to enable reliable large-scale adoption of thermoplastic composites in automotive, aerospace, marine, and construction sectors.

Keywords Thermoplastic composites, Advanced processing techniques, Recyclability, Mechanical performance, Applications

1 Introduction

Thermoplastic composites are a category of cutting-edge materials made of a thermoplastic polymer matrix reinforced with fibrous (glass, carbon, aramid, natural and nanofillers) [1]. In recent years, these composites have attracted considerable interest due to their superior mechanical performance, processing versatility, and recyclability. In comparison to thermoset composites, thermoplastic matrices have reversible melting behavior, because they can be reshaped, remolded, and reprocessed without suffering appreciable property degradation [2]. Therefore, they are not just suitable for

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conventional structural uses but also extremely needed in the context of the worldwide sustainability goals. Their lightweight design, chemical resistance, and rapid processing create thermoplastic composites that are perfect for sectors aiming to minimize emissions and material waste. As environmental regulations become more stringent and the need for recyclable materials grows, thermoplastic composites are becoming a viable solution at the intersection of performance and sustainability [3].

The use of thermoplastic composites is growing in several industries such as electronics, healthcare, automotive, and aerospace because of their exceptional durability, lightweight, and adaptable design [4], [5]. Figure 1 demonstrates the positive characteristics of thermoplastic composite materials [6].

Thermoplastic composites provide design flexibility and robustness, driving advancements in electronics, medical devices, and sporting goods. These composites are being utilized in the automotive industry to substitute metal and thermoset parts, contributing to vehicle light-weighting and enhanced fuel efficiency [7]. Developments in processing technologies like automated tape placement, in-situ consolidation, and additive manufacturing have enabled them to move from specialized to scalable applications and high-caliber production.

The purpose of this review article is to objectively examine the most recent advancements in thermoplastic composite materials, paying special attention to processing methods, recycling pathways, and performance-related issues. It investigates unique production methods that improve productivity and material integrity, as well as discussing



Fig. 1 Illustration of the advantageous features of thermoplastic composite materials [6]

the restrictions associated with long-term durability, interfacial bonding, and mechanical performance under varied loading conditions. The study also explores the role of these materials in the evolution toward a circular economy, emphasizing recycling techniques and life cycle considerations. Through the identification of existing research gaps and industry barriers, the review article explores future research directions and provides information on how these materials can be engineered to accommodate the future generation's sustainability and performance standards.

2 Classification and constituents of thermoplastic composites

2.1 Types of thermoplastic matrices

Thermoplastic composites act as the continuous phase in a composite system, binding the reinforcements and transferring loads. The most widely utilized thermoplastic matrices in composite creation comprise:

2.1.1 Polypropylene (PP)

Polypropylene (PP) is a broadly used thermoplastic polymer categorized by an elevated volume-to-weight ratio, which presents difficulties for landfill management because of its size [8–10]. Despite its glossy look, high tensile strength, and desirable optical characteristics, it makes it a need for packaging polymer [11]. Tertiary carbons present along the PP backbone are more likely to degrade due to shear [12], adds a more reactive methine functional group, which makes it more flammable [13]. PP molecular weights decrease with repeated recycling because of thermo-mechanical and thermo-oxidative chain scission. The degree of crystallinity rises as a result of this deterioration, thus increasing the elastic modulus but decreasing elongation at break [14].

2.1.2 Polyamide (PA)

Polyamide (nylon) was initially introduced into the market by DuPont in 1938, such as toothbrush filaments of polyamide 6,6 (PA66). It is used as a plastic produced in the 1950s as the original engineering plastic, indicating today one of the major and most vital materials of this class [15]. This is supported by its intriguing qualities, such as better chemical resistance, moderately high tensile strength, and stiffness [16].

2.1.3 Poly-ether-ether-ketone (PEEK)

A high-performance thermoplastic composite, PEEK from a polyaryletherketone (PAEK) family, has found extensive application as a biomaterial for spinal, trauma, and orthopedic implants [17]. Several mechanical tests and in vivo simulation investigations (including corrosion and degrading damage) have shown that PEEK is suitable for use in healthcare applications such as orthopedics [18] and trauma as seen in Fig. 2.

2.1.4 Poly-ether-ketone-ketone (PEKK)

Similarly, to PEEK, PEKK also has exceptional mechanical properties, strong resistance to a range of chemicals and environmental factors, and resilience to high temperatures. It is therefore ideally suited for uses involving heavy mechanical loads [19]. Thus, it is employed in several essential aviation parts, comprising thermal shields, brackets, and additional structural components [20].

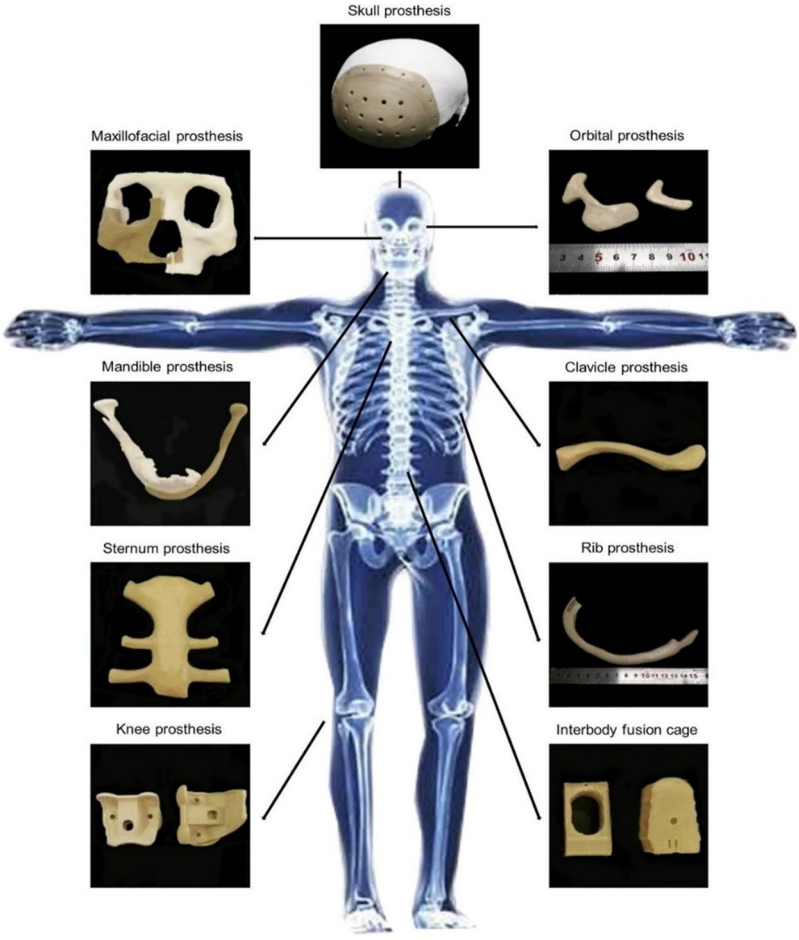


Fig. 2 Schematic demonstrating the applications of PEEK implants in orthopedics [18]. (Open access, CC BY-NC-ND 4.0)

2.2 Reinforcement materials

The reinforcement phase in thermoplastic composites presents load-bearing capacity, stiffness, and strength. The selection of reinforcement is based on the desired thermal, mechanical, or environmental performance.

2.2.1 Carbon fibers

Carbon fibers (CFs) are extensively employed as reinforcements in advanced composite materials due to their high specific strength, high modulus, and excellent wear resistance. Based on fiber length, CFs are categorized into continuous, long, and short carbon fibers (SCFs). Carbon fiber–reinforced polymers combine superior mechanical performance with low density, making them widely used in aerospace, transportation, and sporting goods applications [2].

2.2.2 Glass fibers

Glass fibers are a common type of polymer reinforcement and are used to strengthen the FRPs. Structural composites, printing boards, and a variety of other specialty products are all manufactured with glass fibers [21]. GFRP composites are commonly used in several industrial processes for a variety of reasons [22]. Early Egyptians used GFs

derived from heat-modified glass to make jars. In the 1930s, continuous GFs were first developed for high-strength applications [23]. High strength, flexibility, superior electrical insulation, low heat conductivity, inflammability, rigidity, and protection against chemical damage are the distinctive characteristics of glass fibers [24].

2.2.3 Aramid fibers

Aramid fibers are linear polyamides that do not undergo significant shrinkage during melt processing and exhibit excellent strength and toughness, making them widely used in high-impact applications such as ballistic protection. Although more expensive than glass fibers, aramid fibers (Kevlar) are considered cost-competitive when evaluated in terms of mechanical recyclability [25], [26]. Accordingly, they are attractive reinforcements for thermoplastic composites. Thermoplastics reinforced with aramid fibers have been reported to exhibit outstanding impact resistance. However, their strength and elastic modulus are generally lower than those of glass- or carbon-fiber-reinforced plastics [25]. This behavior is largely attributed to the high flexibility of aramid fibers, which tend to disperse in a deflected or curved state during molding. Furthermore, the strong anisotropy in their mechanical properties between axial and radial directions can limit effective load transfer, leading to suboptimal reinforcement efficiency [25].

2.2.4 Natural fibers

In recent decades, natural fibers have gained considerable attention as alternative reinforcements in polymer composites due to their advantages over conventional glass and carbon fibers. Commonly used natural fibers include flax, hemp, jute, sisal, kenaf, coir, kapok, banana, and henequen [27]. Compared with synthetic fibers, natural fibers offer benefits such as low cost and density, competitive specific tensile properties, non-abrasiveness to processing equipment, reduced skin irritation, lower energy consumption during production, minimal health risks, and enhanced renewability, recyclability, and biodegradability [27–30].

2.2.5 Nanofillers

Carbon nanotubes (CNTs) have attracted significant interest due to their unique structures and exceptional properties [31], including extremely high Young's modulus along the nanotube axis, tunable electrical conductivity ranging from insulating to metallic, and hollow morphology. Graphene nanosheets, with a Young's modulus of about 1 TPa and an ultimate strength approaching 130 GPa, are among the strongest materials known [32]. Clay nanofillers naturally occur as platelets composed of a few to several thousand stacked layers, with a single montmorillonite (MMT) sheet exhibiting an in-plane Young's modulus in the range of 178–265 GPa [32]. These outstanding properties make nanofillers highly effective reinforcements for polymer matrices. Among them, MMT is the most widely used clay nanofiller, consisting of an alumina octahedral sheet sandwiched between two silicate layers [32].

3 Processing technologies

3.1 Compression molding

In general, the compression molding process is widely used to create composite and polymer products [33]. This molding method saw significant expansion in the early

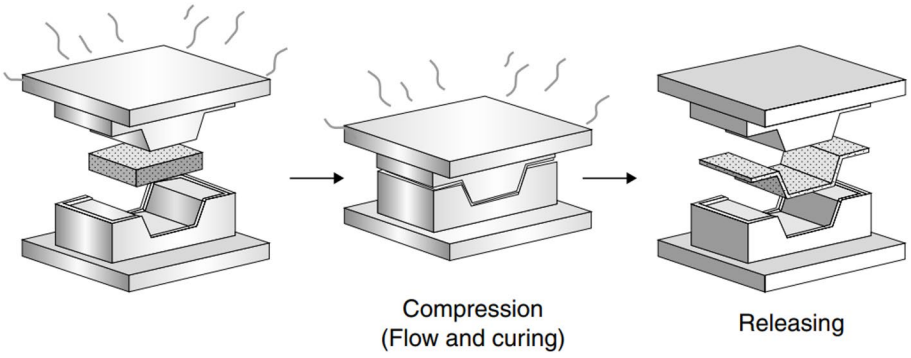


Fig. 3 The compression molding stages [36]. (Elsevier permission number – 6175670512949)

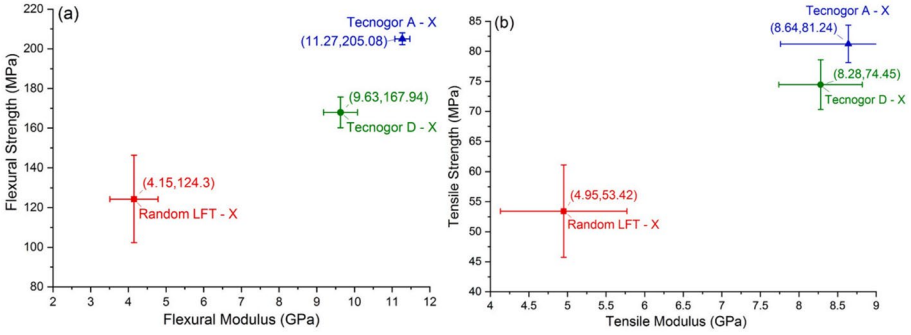


Fig. 4 Mechanical characteristics of Tecnogor A, Tecnogor D, and random LFT samples; **a** flexural strength versus flexural modulus, and **b** tensile strength versus tensile modulus [37]. (Elsevier permission number – 6175670787879)

1950s, especially in automotive applications, since sheet molding materials had been developed [34]. During the compression molding method, the appropriate amount of material is placed between the upper and lower molds. After that, the material turns into the final shape through pressing the mold [35], [36] as depicted in Fig. 3.

Alwekar et al. [37] investigated the effects of processing methods on the microstructural and mechanical characteristics of long fiber thermoplastic (LFT) composites. The results showed that Tecnogor composites, treated with aligned fiber extrusion, perform better than random LFT composites both structurally and mechanically. Figure 4 exhibits excellent tensile and flexural characteristics in Tecnogor composites, attributed to fiber alignment. Whereas, Fig. 5 indicates the decreased and consistently dispersed porosity in Tecnogor samples, verifying improved structural integrity in comparison to random LFT composites.

Moreover, other studies primarily concentrated on the effect of fiber length reduction and distribution during conventional injection molding or extrusion [38–40], not having sophisticated process control or estimate modeling. In comparison to the Tecnogor research, fiber alignment methods, sophisticated imaging, and mathematical modeling methodologies are not integrated in this examination to demonstrate unambiguous relationships between processing techniques as well as mechanical performance. Furthermore, they provide only a few insights into fiber orientation quantification or porosity control, which are significant in accomplishing better mechanical characteristics.

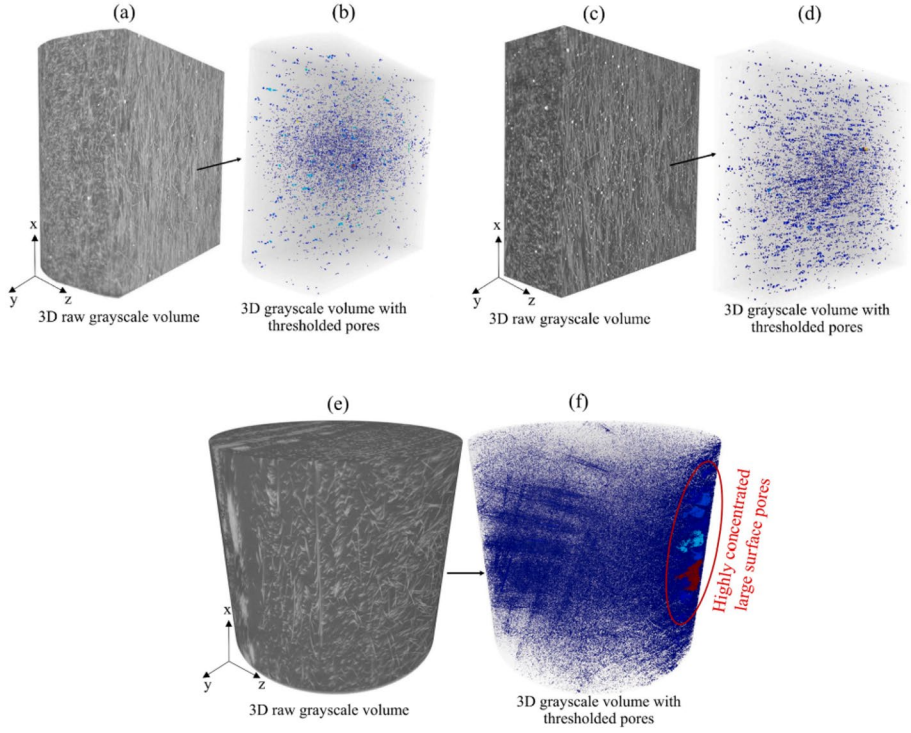


Fig. 5 3D raw grayscale volume and volume with an applied threshold range of pores showing size, shape, and location of pores in (**a**, **b**) Tecnogor A, (**c**, **d**) Tecnogor D, and (**e**, **f**) random LFT samples. The random LFT sample demonstrates a high level of concentration of pores near the surface [37]. (Elsevier permission number – 6175671108530)

3.2 Thermoforming

A generic word for a process that begins with an extruded sheet of plastic is known as thermoforming. The procedure includes heating the plastic sheet to a temperature range at which it becomes soft or malleable [41]. D`Emilia et al. [42] evaluates the shear deformation occurring in glass fiber-reinforced thermoplastic composites using the thermoforming method. It demonstrates robust arrangement between simulation and experimental data, verifying the finite element model and checking the performance of optical systems for determining textile shear deformation. The other investigations, though promising, did not attain the same level of practical accuracy and justification [43–46] as the present work [42]. They mostly depend on controlled testing as well as theoretical modeling without completely capturing the real-time procedure behavior, while this examination effectively integrates experimental data with finite element simulation to generate outcomes that closely resemble real-world production circumstances [42].

3.3 Automated tape placement (ATP)

Automated tape placement is a cutting-edge manufacturing process that creates composite structures by carefully applying continuous fiber tows (usually glass or carbon fibers that have already been resin-impregnated) onto a mold or tool surface in a predetermined pattern [47]. Fontes et al. [48] investigated the utilization of a theory-guided neural network (TgNN) to simulate the three-dimensional temperature distribution through in-situ AFP manufacturing of thermoplastic composites.

As depicted in Fig. 6, the overall setup involved pre-processing as well as data collection using embedded thermocouples and sensors. By incorporating physics-based feature transforms, the TgNN increases estimation precision with fewer data and permits real-time process control. However, these three investigations focus on standard feedforward neural networks or basic heat transfer traditions [49–51], lacking the theory-guided modeling as well as physics-incorporated feature engineering of the TgNN technique, which improves precision as well as model generalization. Thus, gives insignificant results compared to the above Fontes et al. [48] research.

3.4 In-situ consolidation

In-situ consolidation is defined as coupling the thermoplastic towpreg consolidation with the placement method [52]. This method removes a second stage of autoclave or hot-press consolidation, thereby decreasing the manufacturing cost and processing phases. In addition to thermoplastics' superior mechanical properties over thermoset composites, such as interlaminar strength and recyclable potential. Because of this, AFP and thermoplastic composites' in-situ production techniques are extremely desirable in a market where thermoset composites predominate [53]. Figure 7 depicts the type of compaction and heating systems utilized for in-situ consolidation.

Pourahmadi [54] examines the interlaminar shear strength (ILSS) of Carbon/PEEK thermoplastic composites manufactured using autoclave re-consolidation and automated fiber placement (AFP) in-situ consolidation. Short-beam shear (SBS) testing demonstrated that AFP-manufactured laminates display a 37% decrease in ILSS as a result of higher void content. Finite element modeling with cohesive zone elements precisely recorded mechanical behavior, verified by the close match of practical and numerical results as seen in Fig. 8.

Although other studies are useful for examining the effects of process parameters on ILSS, they depend mostly on experimental findings without using predictive simulation. Their results are therefore restricted to the specific circumstances that were examined and lack the more extensive generalizability and application provided by experimental or mathematical frameworks [55–58]. In comparison, exploration that combined both practical verification and sophisticated modeling methods, such as cohesive zone modeling in FEM [54], provide deeper understandings and allow for the design and prediction of the behavior of composites in a variety of structural uses. Table 1 exhibits a comparative analysis of advanced thermoplastic composite processing technologies.

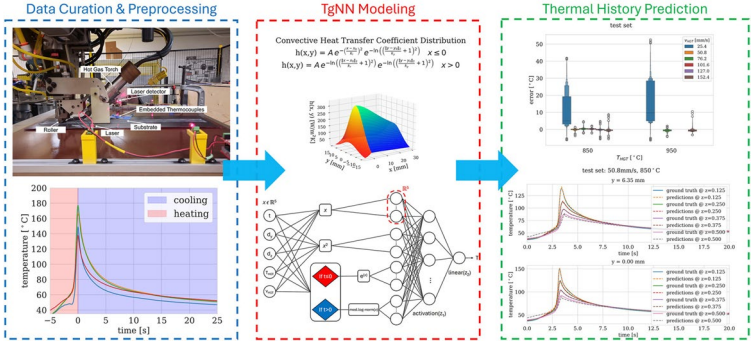


Fig. 6 Workflow of data-driven thermal history estimation in automated fiber placement (AFP) using TgNN modeling [48]. (Open Access CC BY 4.0)

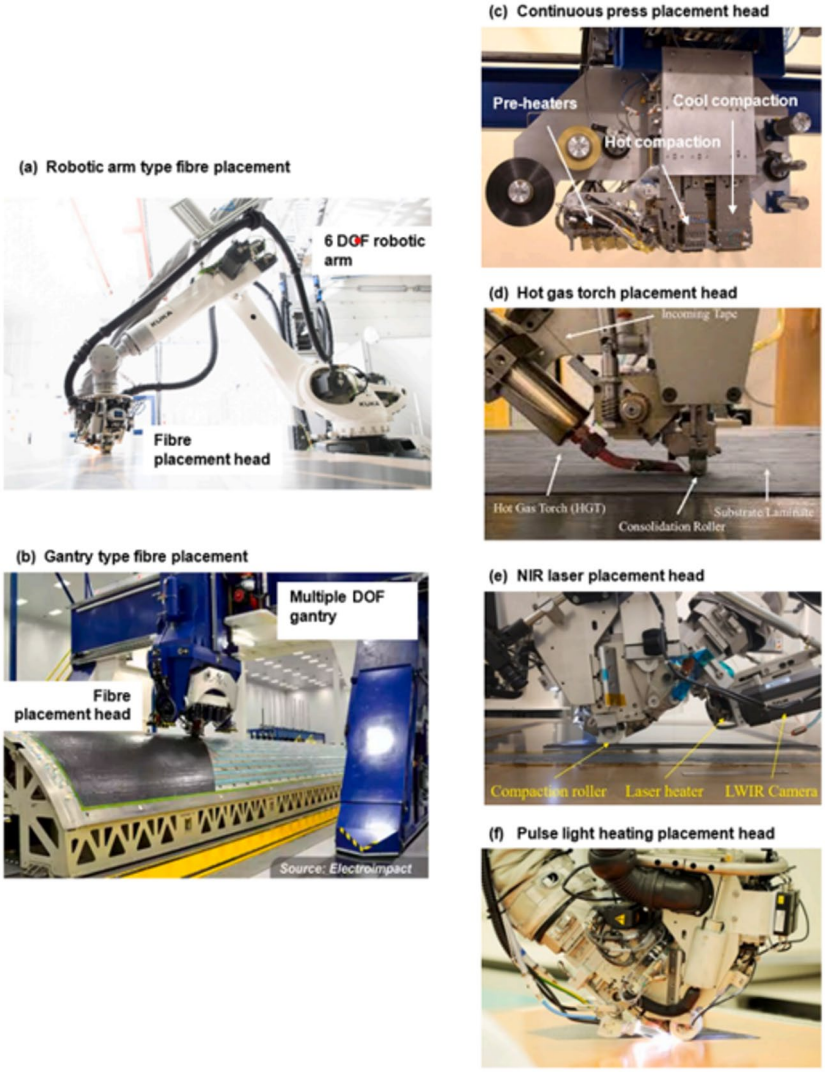


Fig. 7 Automated fiber placement machines with thermoplastic placement heads: **a** robotic arm type, **b** gantry type, **c** heated head automated tape placement (HHATP) developed by Accydyne for NASA, **d** hot gas torch placement head, **e** laser-based placement head by Coriolis, and **f** flash light heating by Heraeus [53]. (Elsevier permission number – 6175671473588)

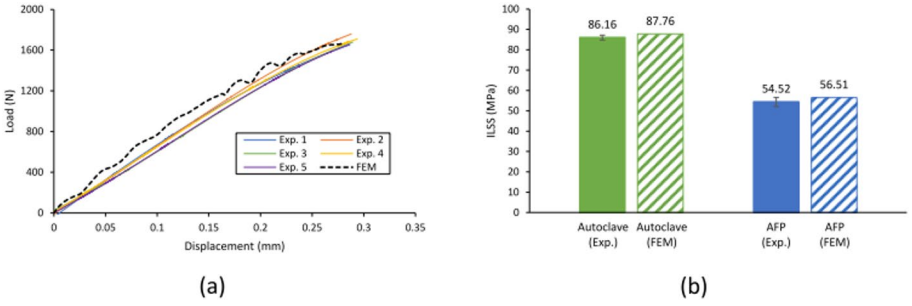


Fig. 8 The response of the CF/PEEK thermoplastic composite during the short-beam shear (SBS) test: **a** load-displacement illustration of reference samples and **b** interlaminar shear strength (ILSS) calculated both numerically and practically [54]. (Elsevier permission number – 6175680123248)

Table 1 Comparison of cutting-edge thermoplastic composite processing technologies based on benefits, drawbacks, and uses

Technique	Advantages	Limitations	Applications	References
Compression molding	Better surface finish and dimension steadiness High replication for large-scale manufacturing	Sophisticated tooling required for complex shapes Insufficient with real-time method monitoring	Automobile structural elements (such as underbody shields and bumpers) Aircraft brackets	[34–36]
Thermoforming	Extremely fast production cycles Lower tooling price, particularly for minimum-volume production	Restricted ability to control the orientation of the fibers during forming Limited to thermoplastics	Lightweight automobile interiors Medical tray holders	[34, 41, 78]
Automated tape placement (ATP)	Allows manufacturing of intricate, large-scale components with minimum physical work High consolidation superiority when integrated with in-situ heating	Very large capital expenditure in robotics systems Flat or slightly curved surfaces are essential for efficient laying.	Wind turbine blades Aerospace structure coverings (fuselage, wing spars)	[47, 48]
In-situ consolidation	Enables continuous production of high-performance thermoplastic composites Consolidation, as well as real-time bonding, decreases energy expenses	Surface roughness may require secondary/additional operations Less resin flow as well as risk of spaces	Marine, rail structural panels Huge composite pipes	[52, 54]

3.5 Liquid composite molding (LCM) processes

Liquid Composite Molding (LCM) processes, particularly Resin Transfer Molding (RTM) and Vacuum-Assisted Resin Infusion (VARI), were used to fabricate fiber-reinforced composite components. In these processes, dry fibrous preforms are first placed into a rigid, closed mold cavity, after which a low-viscosity thermosetting resin is injected or infused under controlled pressure or vacuum conditions. The resin impregnates the reinforcement through porous flow mechanisms governed primarily by Darcy's law, and subsequently cures to form the final composite structure [59].

Before molding, the dry reinforcements were cut and stacked according to the desired laminate architecture and fiber volume fraction. The preforms were then positioned within the mold cavity, and the mold was sealed and clamped to ensure dimensional stability and leak-free operation. Resin injection ports and air vents were strategically located based on mold geometry and permeability distribution to promote uniform flow, minimize race-tracking, and prevent void entrapment [59].

During molding, resin was injected at controlled pressure and temperature conditions to maintain optimal viscosity and ensure complete impregnation of the fibrous preform. Flow front progression was monitored either numerically through mold-filling simulations or experimentally using in-situ sensors, enabling the identification and mitigation of potential dry spots or premature gelation. Once complete mold filling was achieved, the resin was allowed to cure under isothermal or non-isothermal conditions, depending on the resin system, followed by post-curing if required to achieve the target thermomechanical properties [59].

After curing, the mold was opened, and the composite part was demolded, trimmed, and inspected for defects such as voids, incomplete impregnation, or dimensional

inaccuracies. This manufacturing approach enables the production of complex-shaped, high-performance composite components with controlled fiber volume fractions, good surface finish, and relatively low tooling and processing costs compared to autoclave-based methods [60].

Across compression molding, thermoforming, ATP, AFP, and in-situ consolidation processes, common defect mechanisms and their influence on structure–property relationships govern the final performance of thermoplastic composites. Voids and incomplete intimate contact remain among the most prevalent manufacturing defects, arising from high resin viscosity, poor fiber wetting, and uneven pressure distribution, and are known to degrade tensile, impact, and interlaminar shear properties in complex geometries unless suppressed by optimized pressure–temporal modulation and lay-up architectonics during forming operations [61]. Crystallinity development is strongly dependent on cooling rate and thermal history, significantly affecting stiffness, toughness, and dimensional stability, as variations in solidification pathways alter matrix morphology and residual stress profiles. Thermal residual stresses, driven by large mismatches in the coefficients of thermal expansion between fibers and matrix and non-uniform cooling, promote warpage, microcracking, and interfacial debonding, particularly evident in AFP-TPC manufacturing, where warpage deformation has been linked to process-induced stress gradients [62]. Fiber waviness and misalignment, common in draping-intensive thermoforming and automated deposition processes, reduce compressive strength and buckling resistance by introducing local stress concentrations and anisotropic response. Interface quality, dictated by melt viscosity, pressure, surface energy, and consolidation time, governs load transfer efficiency and damage tolerance across all processes, as highlighted in recent work on enhanced manufacturing quality via infrared-assisted AFP that demonstrates improvements in fiber–matrix bonding and reduced defect prevalence [63]. Collectively, these interrelated processing-induced features translate directly into tensile, impact, fatigue, and creep performance, underscoring the need for integrated thermal–mechanical process control strategies rather than isolated optimization of individual manufacturing parameters.

4 Recyclability and sustainability perspectives

4.1 Life cycle assessment (LCA) of thermoplastic composites

Life cycle assessment (LCA) is a technique of analyzing the environmental impact of a product, method, or activity [64], [65]. It determines and measures the amount of energy, materials, and waste released into the environment. It encompasses the full life cycle of an activity, process, or product, comprising: production, shipping, and distribution; conservation, reuse, and use; recycling, and disposal [66]. Hence, it tries to evaluate environmental impacts in all phases of the product life cycle [67], [68] as depicted in Fig. 9. Investigators have carried out LCA research on glass fiber-reinforced bio-based composites [69], natural fiber-reinforced fossil fuel-based polymer composites [70–72], bio-based polymer nanocomposites [73]. Numerous nanofillers, i.e., carbon nanoparticles, nano-clays, and cellulose nanoparticles, have also been investigated [74–76]. However, a thorough analysis of cytotoxicity and genotoxicity is necessary to assess biocompatibility and the health effects of nanocomposites [77].

The cradle-to-gate LCA shows that bio-based unsaturated polyester resin (UPR) reduces climate change impact by ~ 75% at the resin level and by ~ 31% at the composite

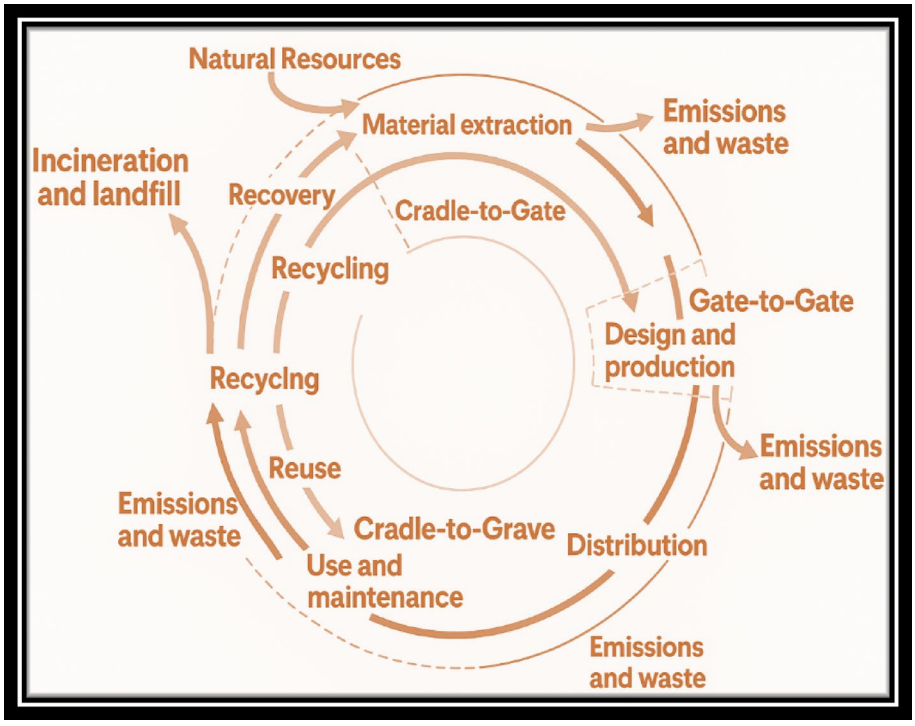


Fig. 9 Life cycle assessment framework [68]. (Elsevier permission number – 6175680553337)

(GFRP) level compared with conventional oil-based systems [69]. Significant reductions are also observed in ozone depletion (48–73%) and fossil resource use (17–35%) [69]. However, impacts related to acidification and eutrophication increase by approximately 25–75%, highlighting trade-offs across impact categories [69]. Single-score results confirm the superior environmental performance of bio-based UPR composites despite these category-specific drawbacks.

Another LCA study shows that PLA has a substantially lower GWP (~0.6 kg CO_2/kg) than conventional plastics such as HDPE (1.8–1.9), PP (1.6–1.9), and PET (2.0–2.4 kg CO_2/kg), mainly due to biogenic carbon uptake during feedstock growth [73]. Product manufacturing and transport contribute moderately, with slightly higher impacts for PET and PLA-based materials due to higher processing temperatures and density [73]. End-of-life impacts are lowest for PLA and PLA nanocomposites when composting is considered, though results improve further if recyclability is included. The addition of nanoclay fillers provides only marginal reductions in overall GWP and energy demand [73].

4.2 Environmental impact and sustainability

The sustainability and environmental effects of thermoplastic composites are crucial factors in both their development and application. Many significant components have emerged and are listed below.

4.2.1 Utilizing renewable resources and lowering carbon footprint

Thermoplastic composites are being created more frequently using bio-based polymers [79–81] and natural fiber reinforcements [82], [83], which would considerably reduce dependence on petroleum-derived materials. By integrating renewable resources,

including natural fibers such as flax, hemp, jute, Chamaerops humilis rachis, palm waste Washingtonia, with polylactic acid (PLA), polyhydroxyalkanoates (PHA), and high-density polyethylene, producers have the potential to reduce the environmental impact of conventional thermoplastics [84–88]. However, the utilization of renewable feedstocks helps to reduce the overall carbon footprint. As they developed, natural fibers absorb carbon dioxide, partially compensating for the emissions produced during the creation of the composite [89]. It can be recycled as well as reprocessed more easily than thermosets, reducing greenhouse gas emissions during the span of a product's life [90].

4.2.2 Embracing biodegradability and diverse end-of-life choices

Sustainability in thermoplastic composites encompasses responsible end-of-life management in addition to manufacturing. Numerous bio-based thermoplastics, including PLA, provide natural biodegradability in industrial composting settings, reducing the long-term buildup of waste in landfills [91], [92]. Figure 10 illustrates the workflow of product development in the case of the production of thermoplastic composites [93]. Furthermore, to biodegradability, thermoplastic composites encourage a range of end-of-life practices, i.e., energy recovery, chemical recycling, and mechanical recycling. Through the conversion of polymers back into monomers for reuse, chemical recycling techniques like depolymerization further improve the recyclability of high-performance bio-composites. These adaptable disposal choices support the ideas of the circular economy, encouraging resource efficiency and minimizing environmental impact over the duration of the product life [94], [95].

4.3 Recycling approaches for thermoplastic composites

The recyclability of thermoplastic composites represents a major advantage over thermoset-based systems, owing to their melt-processability and potential for closed-loop material recovery. However, practical recycling performance depends strongly on fiber type, matrix chemistry, reinforcement architecture, and contamination levels. Recycling strategies for these composites can broadly be classified into mechanical reprocessing and remelting, dissolution/solvent-based routes, and thermochemical processes, each

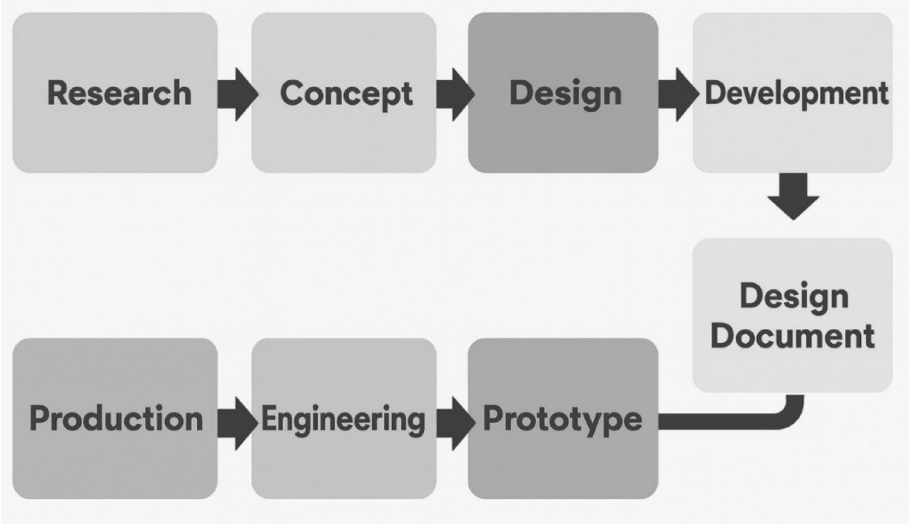


Fig. 10 Steps of sustainable product development from thermoplastic composites [93]. (Open Access - CC BY 4.0)

with distinct advantages and limitations in terms of reclaimed fiber quality, polymer property retention, and environmental performance.

Mechanical recycling is the most established route for thermoplastic composites and involves shredding, remelting, and reprocessing into new parts via extrusion, injection molding, or compression molding. This approach leverages the re-meltability of thermoplastic matrices, enabling recovery and reuse of composite waste, but often results in fiber shortening and polymer degradation that reduce strength and impact resistance in second-generation materials; such effects have been documented in carbon fiber–reinforced PA6 and glass fiber–reinforced PA6 systems processed via mechanical recycling and remolding [96], [97].

Thermochemical recycling methods, such as pyrolysis and gasification, decompose the polymer matrix to recover fibers and are mainly applied to thermoset composites, with limited use in thermoplastics, especially in mixed-waste streams. Although carbon fibers can retain much of their tensile strength, surface degradation reduces interfacial performance, while glass fibers suffer significant strength loss. Consequently, these routes are primarily suitable for contaminated or end-of-life composites where polymer recovery is impractical, and fiber recovery or energy valorization is the main objective [98], [99].

Reclaimed fiber quality depends strongly on the recycling route, i.e., mechanical recycling shortens fibers and limits reuse to short-fiber applications, dissolution preserves fiber length for high-performance reuse, and thermochemical methods retain fiber strength but degrade surface functionality unless re-treated. Repeated reprocessing also degrades thermoplastic matrices through chain scission and crystallinity changes, reducing toughness and strength, particularly in commodity polymers, while high-performance matrices show better but still gradual property loss [100–102].

Although thermoplastic composites offer superior recyclability compared to thermosets, practical challenges remain. Key limitations include contamination in post-consumer waste streams, inconsistent fiber architectures, degradation during repeated thermal cycling, and the lack of standardized recycling infrastructures for continuous-fiber composites. Mechanical recycling, while scalable, is best suited for downcycling applications, whereas dissolution-based methods enable higher-value recovery but require significant process optimization and environmental controls.

Future developments are expected to focus on solvent systems with lower toxicity and energy demand, fiber-compatible re-sizing strategies, and hybrid recycling schemes combining mechanical pre-processing with selective dissolution. Digital product passports, mono-material design strategies, and improved sorting technologies are also anticipated to enhance closed-loop recycling feasibility for thermoplastic composite structures, particularly in automotive and aerospace sectors.

5 Performance and durability challenges

Thermoplastic composites are being rapidly adopted across various industries, but they still face challenges related to performance and durability for long-term structural and semi-structural applications. The challenges are specifically associated with stability under the complex loading, degradation associated with environmental exposure, and weak interfacial bonding between the constituents.

5.1 Mechanical performance under dynamic loading

Thermoplastic composites are increasingly exposed to dynamic service conditions such as cyclic fatigue, impact loading, vibration, and crash-type events, particularly in automotive, aerospace, and marine applications [5], [103], [103–106]. Under such loading conditions, these materials commonly exhibit stiffness degradation, matrix microcracking, progressive fiber-matrix debonding, and eventual fiber fracture [107], [108]. Compared with thermoset composites, thermoplastics generally offer superior damage tolerance and impact resistance. However, their viscoelastic nature can accelerate damage accumulation under long-term dynamic loading. Experimental fatigue and creep tests, including hold time and cyclic creep protocols, demonstrate that viscoelastic creep, stress relaxation, self-heating, and mean stress effects significantly shorten fatigue life and promote creep-type failure during cyclic loading [109]. Similarly, static, creep, and fatigue tensile experiments on polypropylene-based thermoplastic composites reveal that matrix damage accumulation governed by viscoelastic mechanisms controls the overall fatigue performance [110]. Tensile relaxation and long-term creep studies further show that viscoelastic and viscoplastic flow, combined with slow microcrack growth, are time-dependent processes that dominate lifetime behavior and can ultimately lead to premature failure [111].

Hybrid thermoplastic composites, although promising in improving toughness, sometimes demonstrate complex crack propagation mechanisms due to heterogeneous stress distribution. High strain-rate loading, such as impact and crash scenarios, can further induce localized plastic deformation, delamination, and interlaminar failure, reducing the structural integrity of the component [112]. Experimental fracture tests on woven ply hybrid fiber thermoplastic laminates have revealed simultaneous fiber breakage, local plastic deformation, delamination, and complex crack growth paths during loading [113]. In addition, hybrid thermoplastic composites such as carbon flax hybrids in a polyamide matrix show altered interlaminar bonding and distinct failure behaviour compared with non-hybrid systems, indicating a heterogeneous stress response at the fiber-matrix interfaces that strongly influences crack initiation and propagation [114]. Studies on hybrid polymer composites under high strain rate and impact loading further highlight that hybridization modifies dominant failure mechanisms, including delamination and matrix plasticity, when compared with single-phase composites under dynamic service conditions [115].

5.2 Interfacial bonding and fiber–matrix adhesion issues

Interfacial bonding between reinforcing fibers and thermoplastic matrices plays a decisive role in stress transfer efficiency and the overall mechanical performance of thermoplastic composites [116]. Weak fiber-matrix adhesion has been widely identified as one of the primary causes of premature failure in thermoplastic composites, particularly under tensile loading conditions [117]. During multiaxial fatigue loading scenarios, shear damage initiated in all thermoplastics, regardless of whether they were under notched or unnotched conditions. However, under in-phase axial torsion loading, crack propagation in both specimen types occurred predominantly in a tensile mode [118].

Natural fibers are inherently hydrophilic due to their polar hydroxyl groups, while common thermoplastic polymer matrices are hydrophobic, leading to poor compatibility, weak interfacial bonding, and reduced mechanical performance in composites

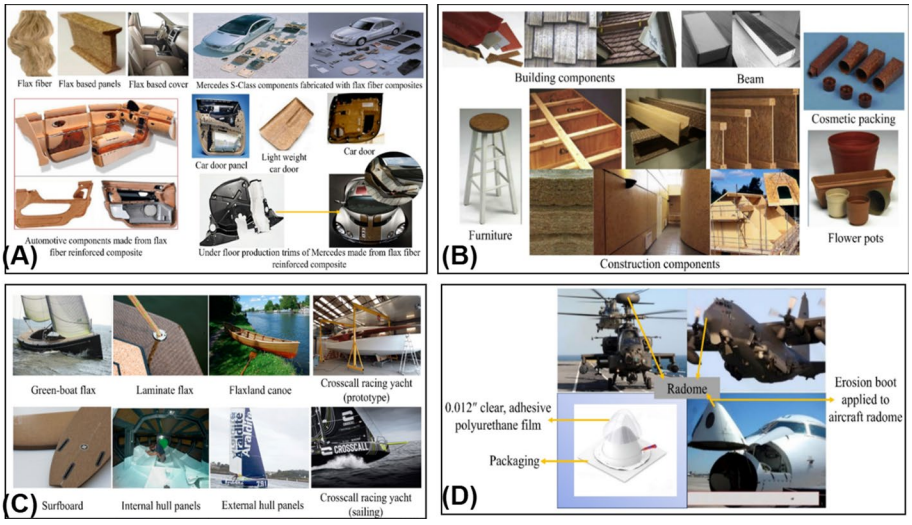


Fig. 11 Thermoplastic composites utilized in various parts of: **a** automotive components [128], [129], [132], [134]; **b** building and construction component [128], [134]; **c** ship and fishing boats [138]; **d** aircraft radome [132]. (118, 123, 126, 130-Open Access CC BY 4.0; 120- John Wiley and Sons permission number: 6175681167556)

unless surface modifications or compatibilizers are used [119]. This mismatch results in interfacial debonding, moisture uptake, and accelerated degradation during environmental exposure [120], [121]. Surface treatments and coupling agents such as silanes and maleic-anhydride-based compatibilizers have been introduced to improve interfacial strength; however, uniform distribution and long-term stability of these treatments remain challenging [122], [123].

Residual stresses develop in thermoplastic composites because the matrix and fibers cool from processing temperatures with different thermal expansion values. This mismatch at the fiber–matrix interface creates internal stresses that influence mechanical behavior and can contribute to damage development [124]. Review studies on thermoplastic composites show that the mismatch in thermal expansion between fibers and matrix leads to residual tensile and compressive stresses in the constituents after cooling, and these stresses influence interfacial strength, debonding, and cracking upon thermal cycling [125]. Hygrothermal analyses highlight that differential expansion from temperature changes and moisture content creates internal stresses that contribute to matrix cracking and fiber–matrix debonding [126].

Addressing these interfacial challenges requires advanced surface modification strategies, optimized fiber sizing techniques, and improved processing control to ensure effective wetting, reduced void content, and enhanced long-term durability of thermoplastic composite structures.

6 Applications of thermoplastic composites

6.1 Automotive and aerospace industry

Thermoplastic composites are being utilized more and more in the automotive sector because of their exceptional recyclability, great impact resistance, and lightweight nature [127]. The lightweight advantages of thermoplastic composites lead to improved vehicle fuel efficiency [128]. They are used in the production of interior parts such as under-the-hood parts, instrument panels, seat frameworks, and car door panels [129] as depicted in Fig. 11. Besides internal use in automobiles, synthetic polymer NFRCs have been used

outside, including the area between the headlight, as well as vehicles of passenger buses [130]. Whereas, polymer matrix composites are commonly utilized in aerospace applications for emerging lightweight buildings [131], [132]. Thermoplastic composites are employed in the construction of various airplane sections [60]. Because they are less dense than synthetic fiber-based composites, more weight can be handled by the airplane's structure [133].

6.2 Construction and building

Thermoplastic composites have been widely used in construction since prehistoric times. Mostly, composites used as bricks in home construction are made of sawdust, husk, and short straw fibers integrated with clay [134]. These composites provide several benefits in building, such as biodegradability, rot resistance, sufficient strength, low maintenance, and light weight. [135, 136]. However, flax fiber composites are used as building materials in conjunction with concrete structures to improve the structural strength [137].

6.3 Ship and fishing boats

Laminated fiber-reinforced composite materials are used today in every aspect of the marine sector. These materials are widely utilized in labor and fishing boats, the building of fast ferries, racing yachts, and pleasure boats [131]. This is because using composite materials has several advantages over using steel, aluminum, or wood. Important advantages include simplicity in creating intricate shapes, corrosion resistance [138].

7 Conclusion and prospects

Thermoplastic composites are a transformative class of materials that effectively combine mechanical performance, lightweight design, and recyclability, making them highly attractive for next-generation structural and semi-structural applications. This review highlights significant progress in thermoplastic composite matrices, reinforcements, and advanced processing technologies, demonstrating their increasing industrial relevance. Manufacturing methods such as automated tape placement and in-situ consolidation offer clear benefits in productivity and structural integration, while recycling and life cycle assessment studies emphasize their potential contribution to sustainable material systems.

However, several critical challenges still limit widespread adoption. Long-term durability under cyclic, impact, and hygrothermal loading remains a major concern due to viscoelastic deformation, interfacial degradation, and residual stress development. Weak fiber-matrix adhesion, moisture-induced damage, and processing-induced voids significantly affect fatigue resistance and service life. Addressing these issues requires improved interfacial engineering strategies, optimized surface treatments, and better control of processing parameters to ensure steady consolidation and long-term stability.

Future research should focus on multi-scale experimental and modeling approaches to connect processing, microstructure, and performance [139]. Integrating physics-informed machine learning, real-time process monitoring, and sustainable material design will be crucial for developing durable, high-performance thermoplastic composites that meet future industrial and environmental needs [140–142].

Abbreviations

PP	Polypropylene
PA	Polyamide
PEEK	Polyether ether ketone
PAEK	Polyaryletherketone
PEKK	Polyetherketoneketone
CMCs	Ceramic matrix composites
CNTs	Carbon nanotubes
FRP	Fiber-reinforced plastic
ATP	Automated tape placement
TgNN	Theory-guided neural network
LFT	Long fiber thermoplastic
ILSS	Interlaminar shear strength
AFP	Automated fiber placement
SBS	Short-beam shear
LCA	Life cycle assessment
PLA	Polylactic acid
PHA	Polyhydroxyalkanoates

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